

Physics 214/224, Practice Quiz 5

Primary instructor: Yurii Maravin

WID: _____

NAME: _____

Instructions. Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: "On my honor as a student I have neither given nor received unauthorized aid on this academic work."

For two quick points, circle your studio instructor:

Lohman (TU 7:30) Lohman (TU 9:30) Weaver (TU 9:30)

Magrakvelidze (TU 11:30) Maravin (TU 1:30) Weaver (TU 3:30)

Magrakvelidze (WF 7:30) Magrakvelidze (WF 9:30)

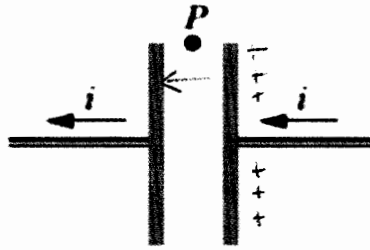
Golin (WF 11:30)

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.



Part I: All questions to be answered on your Scantron card (48 points total)

The figure below shows a parallel-plate capacitor and the current in the connecting wire that is **charging** the capacitor. In the next three questions you need to identify correct statements regarding the electric field and its flux between the capacitor's plates, direction of the displacement current, and the direction of magnetic field.



1. (3 points) Select the correct statement below:

- ☒ (a) The direction of the electric field $\vec{E}$ between the plates is leftward and the electric flux through the vertical plane between the capacitor plates is increasing.
- (b) The direction of the electric field $\vec{E}$ between the plates is rightward and the electric flux through the vertical plane between the capacitor plates is increasing.
- (c) The direction of the electric field $\vec{E}$ between the plates is leftward and the electric flux through the vertical plane between the capacitor plates is decreasing.
- (d) The direction of the electric field $\vec{E}$ between the plates is rightward and the electric flux through the vertical plane between the capacitor plates is decreasing.

2. (3 points) Select the correct statement below:

- ☒ (a) The direction of the displacement current i_d between the plates is leftward.
- (b) The direction of the displacement current i_d between the plates is rightward.
- (c) The displacement current is zero, i.e., no direction.

$$(I_d = \epsilon_0 \frac{d\Phi}{dt})$$

↑
increases
I is along $\vec{E}$

3. (3 points) Select the correct statement below?

- ☒ (a) The magnetic field at point P is directed into the page.
- (b) The magnetic field at point P is directed out of the page.
- (c) There is no magnetic field at point P because there is no actual current of charged particles between the plates.

4. (3 points) Gauss' law for magnetism:

- (a) can be used to find $\vec{B}$ due to given currents, provided there is enough symmetry
- (b) is false because there are no magnetic charges
- (c) can be used with open surfaces because there are no magnetic charges
- (d) contradicts Faraday's law because one says $\Phi_B = 0$ and the other says $\mathcal{E} = -d\Phi_B/dt$.
- ☒ (e) none of the above

5. (3 points) Suppose you are looking into one end of a long cylindrical tube in which there is a uniform electric field, pointing away from you. If the magnitude of the field is decreasing with time the direction of the induced magnetic field is:

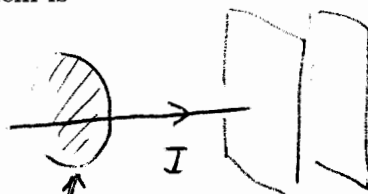
- ☒ (a) clockwise
 (b) toward you
 (c) counterclockwise
 (d) away from you
 (e) to your right

$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$ direction of current is along $\Delta \vec{E}$ - toward you.
 use RHR, so counterclockwise



6. (3 points) A current of 1 A is used to charge a parallel plate capacitor with square plates. If the area of each plate is 0.6 m^2 , the displacement current through a 0.3 m^2 area wholly between the capacitor plates and parallel to them is

- (a) 0.5 A
 (b) 0.25 A
 (c) 0.7 A
 (d) 1 A
 (e) 2 A



$\vec{B}$ is produced by $I = 1 \text{ A}$ here is using flat disk or I_d if using capacitors, so $I_d(\text{full}) = 1 \text{ A}$ as well, and $I_d(0.3 \text{ m}^2) = \frac{1}{2} \text{ A}$

7. (3 points) Select the correct statement:

- (a) blue light has a higher frequency than X rays
 (b) radio waves have a higher frequency than gamma rays
 (c) ultraviolet light has a longer wavelength than infrared
 (d) gamma rays have a higher frequency than infrared waves
 (e) electrons are a type of electromagnetic wave

8. (3 points) Consider radio waves (r), visible light (v), infrared (i), X rays (x), and ultraviolet (u). In order of increasing frequency, they are:

- (a) i, r, v, u, x
 (b) x, u, v, i, r
 (c) r, v, i, x, u
 (d) r, i, v, u, x
 (e) r, i, v, x, u

r i v u x

9. (3 points) Radio waves differ from visible light waves in that radio waves:

- (a) have a higher frequency
 (b) travel faster
 (c) travel slower
 (d) have a lower frequency
 (e) require a material medium to propagate

10. (3 points) Which of the following is NOT true for electromagnetic waves?

- (a) they transport energy
- (b) they transport momentum
- (c) they consist of changing electric and magnetic fields
- ☒ (d) they travel at different speeds in vacuum, depending on their frequency
- (e) they can be reflected

11. (3 points) The sun is about 1.5×10^{11} m away. The time for light to travel this distance is about:

- (a) 4.5×10^{19} seconds
- (b) 8 seconds
- (c) 8 hours
- ☒ (d) 8 minutes
- (e) 8 years

$$\frac{1.5 \cdot 10^{11} \text{ m}}{3 \cdot 10^8 \text{ m/s}} = 0.5 \cdot 10^3 \approx 500 \text{ s} \approx 8 \text{ min}$$

12. (3 points) If the electric field in a plane electromagnetic wave is given by $E_m \sin[(3 \times 10^6 \text{ m}^{-1}x) - \omega t]$, the value of ω is:

- (a) 0.01 rad/s
- (b) 10 rad/s
- (c) 100 rad/s
- ☒ (d) 9×10^{14} rad/s
- (e) 9×10^{16} rad/s

$$kx - \omega t = \text{const} \Rightarrow v_p = c = \frac{\omega}{k} \Rightarrow$$

$$\omega = kc = 3 \cdot 10^6 \frac{1}{\text{m}} \cdot 3 \cdot 10^8 \frac{\text{m}}{\text{s}} = 9 \cdot 10^{14} \frac{1}{\text{s}}$$

13. (3 points) A man holds a rectangular card in front of and parallel to a plane mirror. In order for him to see the entire image of the card, the least mirror area needed is:

- (a) that of the pupil of his eye
- (b) one-half that of the card
- ☒ (c) one-fourth that of the card
- (d) an amount which decreases with his distance from the mirror
- (e) that of the whole mirror, regardless of its size



14. (3 points) Which property of a light wave *does not* change as the wave passes from one transparent medium to another:

- ☒ (a) Frequency
- (b) Wavelength
- (c) Phase velocity
- (d) Amplitude

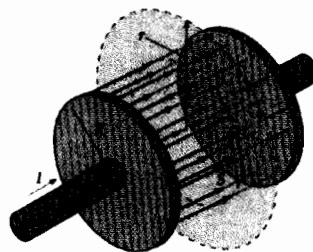
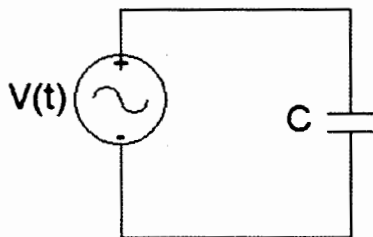
15. (3 points) An object is 2 m in front of a plane mirror. Its image is:

- (a) virtual, inverted, and 2 m behind the mirror
- (b) virtual, inverted, and 2 m in front of the mirror
- (c) virtual, erect, and 2 m in front of the mirror
- (d) real, erect, and 2 m behind the mirror
- ☒ (e) none of the above

16. (3 points) The physical phenomenon that makes fiber optic signal transmission possible is

- (a) Dispersion
- (b) Polarization by reflection
- (c) Small angle refraction
- ☒ (d) Total internal reflection

Part II: Work out this part in the space provided. Show your work!
(50 points total)



17. (16 points) An AC voltage $V(t) = V_0 \cos \omega t$ is applied across a capacitor of capacitance C . The capacitor is made of two circular plates each of radius R .

(a) (2 points) Find the expression for the charge q on the capacitor as a function of time t .

$$q = CV(t) = C V_0 \cos \omega t$$

(b) (2 points) Find the separation d between the capacitor plates in terms of ϵ_0 , C and R .

$$C = \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 \pi R^2}{d} \Rightarrow d = \frac{\epsilon_0 \pi R^2}{C}$$

(c) (3 points) Use the result of (a) and (b) to find the expression for the electric field $E(t)$ between the capacitor plates as a function of time t .

$$E(t) = \frac{V(t)}{d} = \frac{V_0 \cos \omega t}{\epsilon_0 \pi R^2 / C} = \frac{C V_0}{\epsilon_0 \pi R^2} \cos \omega t$$

(d) (3 points) Find the expression for the rate of change of the electric flux $\frac{d\Phi_E}{dt}$ through the circle of radius r centered at the radial axis of symmetry of the capacitor (see the figure above) as a function of time t and position r inside the capacitor for $r < R$.

$$\Phi_E = \pi r^2 E(t) = \frac{C V_0 r^2}{\epsilon_0 R^2} \cos \omega t \Rightarrow \frac{d\Phi_E}{dt} = \frac{C V_0 r^2}{\epsilon_0 R^2} \omega \sin \omega t$$

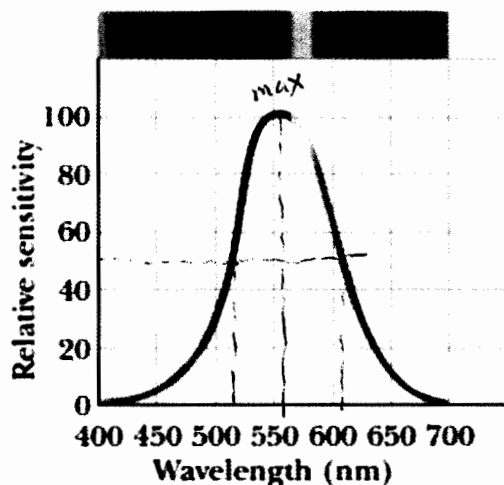
(e) (3 points) Repeat the work above for $r > R$. Assume that the electric field exists only between the plates of the capacitor.

$$\Phi_E = \pi R^2 E(t) \text{ (no field outside of capacitor)} \\ \frac{d\Phi_E}{dt} = \frac{C V_0 R^2}{\epsilon_0 R^2} \omega \sin \omega t = \frac{C V_0}{\epsilon_0} \omega \sin \omega t$$

(f) (3 points) If $V_0 = 180V$, $\omega = 377 \text{ rad/s}$, and $C = 10\mu\text{F}$, what is the peak magnetic field (in Tesla) at the circumference ($r = R$) of the capacitor, where $R = 5.00 \text{ cm}$?

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc} + \mu_0 \epsilon_0 \frac{d\Phi}{dt} \Rightarrow$$

$$2\pi R B = \mu_0 \epsilon_0 \frac{C V_0 \omega}{\epsilon_0} \Rightarrow B = \frac{C V_0 \omega \mu_0}{2\pi R} = 2.7 \cdot 10^{-6} \text{ T}$$



18. (12 points) The figure below shows the relative sensitivity of the average human eye to electromagnetic waves at different wavelengths.

- (a) (2 points) From the figure, approximate the smaller wavelength at which the eye of a standard observer has half the eye's maximum sensitivity.

$$\lambda = 510 \text{ nm}$$

- (b) (2 points) From the figure, approximate the larger wavelength at which the eye of a standard observer has half the eye's maximum sensitivity.

$$\lambda = 610 \text{ nm}$$

- (c) (2 points) What is the wavelength of the light at which the eye is the most sensitive?

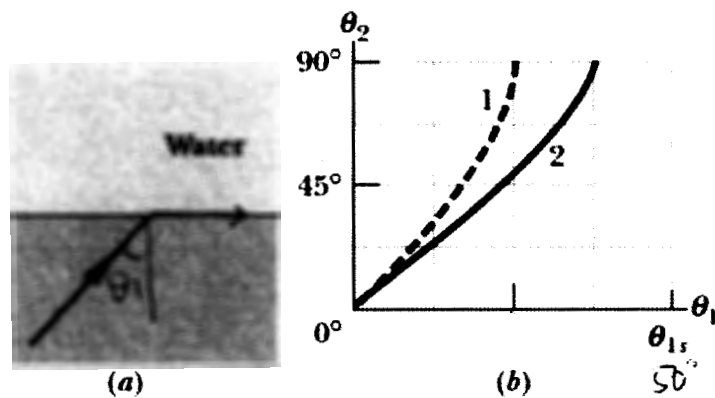
$$\lambda = 560 \text{ nm}$$

- (d) (3 points) What is the frequency of the light at which the eye is most sensitive?

$$f = \left(\frac{\lambda}{c} \right)^{-1} = \frac{3 \cdot 10^8 \text{ m/s}}{560 \text{ nm}} = \frac{3 \cdot 10^8 \text{ m/s}}{560 \cdot 10^{-9} \text{ m}} = 5.36 \cdot 10^{14} \text{ Hz}$$

- (e) (3 points) What is the period of the light at which the eye is the most sensitive?

$$T = \frac{1}{f} = 1.87 \cdot 10^{-15} \text{ sec.}$$



19. (12 points) In figure (a), a light ray in an underlying material is incident at angle θ_1 on a boundary with water, and some of the light refracts into the water. There are two choices of underlying material. For each, the angle of refraction θ_2 versus the incident angle θ_1 is given in figure (b). The horizontal axis scale is set by $\theta_{1s} = 50^\circ$.

- (a) (3 points) Without calculation, determine whether the index of refraction of material 1 is greater or less than the index of water. Explain your reasoning.

At $\theta_1 = 25^\circ$, $\theta_2 = 90^\circ$ Since $n_1 \sin \theta_1 = n_2 \sin \theta_2$
 $n_1 > n_2$

($\theta_2 > \theta_1$, so $n_2 < n_1$)

- (b) (3 points) Without calculation, determine whether the index of refraction of material 2 is greater or less than the index of water. Explain your reasoning.

For $\theta_1 = 37^\circ$, $\theta_2 = 90^\circ \Rightarrow$ similarly to (a), $n_1 > n_2$.

- (c) (3 points) Determine the index of refraction of material 1.

$$n_1 = \frac{n_2 \sin \theta_2}{\sin \theta_1}, \text{ for } \theta_2 = 90^\circ \text{ \& } \theta_1 = 25^\circ$$

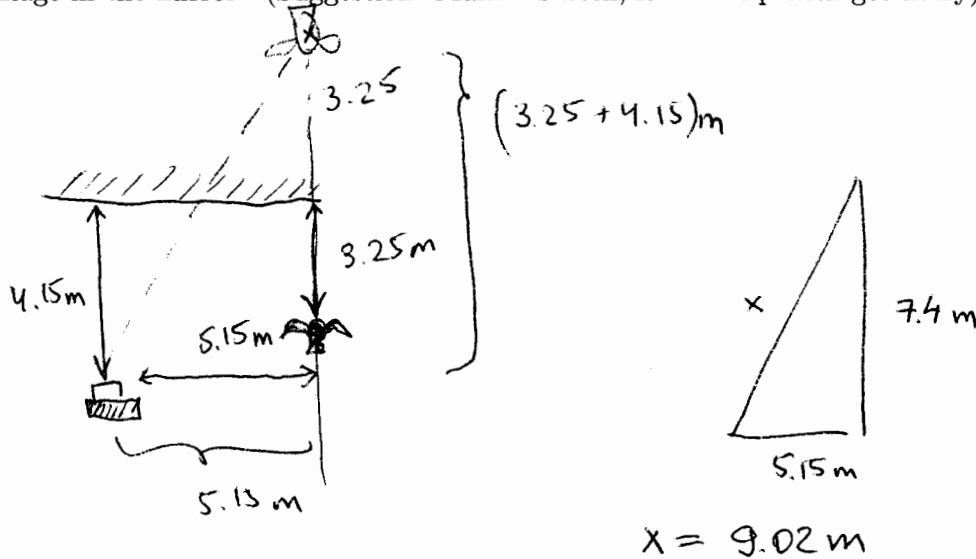
$$n_1 = \frac{1.33}{\sin 25^\circ} \approx 2.32 \text{ } 3.15$$

- (d) (3 points) Determine the index of refraction of material 2.

$$n_1 = \frac{1.33}{\sin 37^\circ} \approx 2.21$$

20. (10 points) You look through a camera toward an image of a hummingbird in a very long plane mirror. The camera is 4.15 m in front of the mirror. The bird is at camera level, 5.15 m to your right and 3.25 m from the mirror.

- (a) (4 points) What is the distance between the camera and the apparent position of the bird's image in the mirror? (Suggestion: Make a sketch, it will help with geometry).



- (b) (4 points) If a bird is moving directly towards the mirror, she will see itself as an image moving towards it. If a speed of a bird is 10 m/s, how quickly is the image moving towards the bird?

The distance between the bird and the image is $L = 2x$, where x is the distance between the bird & mirror. Then

$$v_L = \frac{dL}{dt} = \frac{d(2x)}{dt} = 2 \frac{dx}{dt} = 2 \cdot 10 \text{ m/s} = 20 \text{ m/s}$$

twice as fast!

- (c) (2 points) As you are making a picture of a bird's image in the mirror, you realize that the image must be real and not virtual. Explain why is this reasoning incorrect.

You can make a picture of both virtual & real images. You can project real image on a sheet of paper to do that for example. The image is virtual because real light rays do not form it, light is not converged in the place of an image.